

THE TWENTY-THIRD WILLIAM LOWELL PUTNAM MATHEMATICAL COMPETITION

December 1, 1962

Morning Session

1. Given five points in a plane, no three of which lie on a straight line, show that some four of these points form the vertices of a convex quadrilateral. (page 554)

2. Find every real-valued function f whose domain is an interval I (finite or infinite) having 0 as a left-hand endpoint, such that for every positive member x of I the average of f over the closed interval $[0, x]$ is equal to the geometric mean of the numbers $f(0)$ and $f(x)$. (page 555)

3. In a triangle ABC in the Euclidean plane, let A' be a point on the segment from B to C , B' a point on the segment from C to A , and C' a point on the segment from A to B such that

$$\frac{AB'}{B'C} = \frac{BC'}{C'A} = \frac{CA'}{A'B} = k,$$

where k is a positive constant. Let Δ be the triangle formed by parts of the segments obtained by joining A and A' , B and B' , and C and C' . Prove that the areas of the triangles Δ and ABC are in the ratio.

$$\frac{(k-1)^2}{k^2 + k + 1}. \quad (\text{page 556})$$

4. Assume that $|f(x)| \leq 1$ and $|f''(x)| \leq 1$ for all x on an interval of length at least 2. Show that $|f'(x)| \leq 2$ on the interval. (page 559)

5. Evaluate in closed form

$$\sum_{k=1}^n \binom{n}{k} k^2. \quad (\text{page 560})$$

Note:

$$\binom{n}{k} = \frac{n(n-1) \cdots (n-k+1)}{1 \cdot 2 \cdots k}.$$

6. Let S be a set of rational numbers such that whenever a and b are members of S , so are $a + b$ and ab , and having the property that for every rational number r exactly one of the following three statements is true:

$$r \in S, \quad -r \in S, \quad r = 0.$$

Prove that S is the set of all positive rational numbers. (page 561)

Afternoon Session

1. Let $x^{(n)} = x(x - 1) \cdots (x - n + 1)$ for n a positive integer and let $x^{(0)} = 1$. Prove that

$$(x + y)^{(n)} = \sum_{k=0}^n \binom{n}{k} x^{(k)} y^{(n-k)}. \quad (\text{page 562})$$

Note:

$$\binom{n}{k} = \frac{n(n-1) \cdots (n-k+1)}{1 \cdot 2 \cdots k}.$$

2. Let R be the set of all real numbers and S the set of all subsets of the positive integers. Construct a function f whose domain is R and whose range is in S , such that $f(a)$ is a proper subset of $f(b)$ whenever $a < b$.

(page 563)

3. Let S be a convex region in the Euclidean plane containing the origin. Assume that every ray (that is, half-line) from the origin has at least one point outside S . Prove that S is bounded. (A region in the plane is defined to be convex if and only if the line segment joining every pair of its points lies entirely within the region.)

(page 564)

4. The Euclidean plane is divided into regions by drawing a finite number of circles. Show that it is possible to color each of these regions either red or blue in such a way that no two adjacent regions have the same color. (Two such regions are said to be adjacent if and only if their boundaries have an arc of a circle in common.)

(page 566)

5. Prove that for every integer n greater than 1:

$$\frac{3n+1}{2n+2} < \left(\frac{1}{n}\right)^n + \left(\frac{2}{n}\right)^n + \cdots + \left(\frac{n}{n}\right)^n < 2. \quad (\text{page 566})$$

6. Let

$$f(x) = \sum_{k=0}^n a_k \sin kx + b_k \cos kx,$$

where a_k and b_k are constants. Show that, if $|f(x)| \leq 1$ for $0 \leq x \leq 2\pi$ and $|f(x_i)| = 1$ for $0 \leq x_1 < x_2 < \cdots < x_{2n} < 2\pi$, then $f(x) = \cos(nx + a)$ for some constant a .

(page 567)